

USHNA MALIK

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EDUCATION

University of Iowa

Ph.D., Computer Science

Advisor: Prof. Rishab Nithyanand

Iowa City, IA, USA

August 2025 – Present

Lahore University of Management Sciences (LUMS)

B.Sc. Computer Science, with Distinction

Lahore, Pakistan

August 2021 – May 2025

Stanford University

Undergraduate Summer Semester Candidate

Stanford, CA, USA

June 2023 – August 2023

RESEARCH EXPERIENCE

Modeling AI Intermediaries in Information-Seeking

Advisor: Dr. Rishab Nithyanand

Auditing how AI intermediaries prioritize, frame, and transform content before users see it; designing controlled prompt sets and repeated-measurement protocols to assess source preference and framing, with implications for AI-mediated information access and platform accountability.

Physician–LLM Interaction during Diagnostic Reasoning in an LMIC

CHI 2026

Advisors: Dr. Ihsan Ayyub Qazi, Dr. Ayesha Ali

Led a mixed-methods study of how physicians prompt, verify, and integrate LLM diagnostic suggestions in resource-constrained healthcare settings; developed a category taxonomy of prompting strategies revealing patterns of calibrated trust and human oversight, informing design of clinical AI workflows.

AI-Generated Multimodal Political Persuasion Campaigns

PNAS Nexus (in prep.)

Advisors: Dr. Ihsan Ayyub Qazi, Dr. Ayesha Ali

Led an RCT ($N = 197$) measuring the persuasive impact of LLM-generated, microtargeted political videos in a Global South setting; findings inform debate on AI-enabled influence operations and platform safeguards.

Automation Bias in LLM-Assisted Diagnostic Reasoning

NEJM AI 2026

PI: Dr. Ihsan Ayyub Qazi

Contributed as Research Assistant to an RCT ($N = 44$) examining whether AI-literacy training preserves appropriate reliance when LLM diagnostic suggestions contain errors; findings reveal persistent susceptibility to automation bias even after training, with implications for clinical AI deployment safeguards.

TEACHING EXPERIENCE

University of Iowa

Graduate Teaching Assistant

2025 – 2026

- CS 1110: Introduction to Computer Science. (Fall 2025 and Spring 2026)

Lahore University of Management Sciences

Undergraduate Teaching Assistant

2023 – 2024

- CS334/EE334: Principles and Techniques of Data Science (Fall 2024)
- CS202/EE202: Data Structures (Spring 2024)
- CS210/MATH252: Discrete Mathematics (Fall 2023)

LUMS Learning Institute

August 2024 - Dec 2024

Teaching Assistant, Health Data Science (Quantitative Techniques) Module

Trained practicing health professionals in predictive analytics and exploratory data analysis as part of the Certificate in Health Professions Education (CHPE) program.

Stanford University

June 2023 - August 2023

Student Service Provider, Office of Accessible Education

Adapted lecture materials for COMM 102S to meet accessibility requirements.

ACHIEVEMENTS

Graduated with Distinction, LUMS

2025

Awarded for sustained academic excellence across the BS Computer Science program.

Dean's Honor List, LUMS

Fall 2021 - Spring 2025

Placed on the Dean's Honor List for 4 consecutive years, representing the top 10% of the batch.

TECHNICAL SKILLS

Programming: Python, C, C++, Haskell, TypeScript.

Data and ML: pandas, scikit-learn, FAISS; statistical analysis in R.

Web and Mobile: HTML, CSS, JavaScript, React, Flutter, Dart.

Tooling: Git, L^AT_EX, MySQL, Selenium.

Multimodal Stack: ElevenLabs, Midjourney, Synthesia, Flicki.

RELEVANT COURSEWORK

University of Iowa: Topics in Social Computing Systems; Computer Security; Design and Analysis of Algorithms; Theory of Computation.

Lahore University of Management Sciences:

Graduate-level: Machine Learning; Topics in Large Language Models; Human-Computer Interaction and Design; Topics in Internet Research.

Undergraduate-level: Artificial Intelligence; Principles and Techniques of Data Science; Software Engineering; Databases; Advanced Programming.

Stanford: High Performance Computing; Deception and Technology.

SERVICE AND MENTORING

Peer Ambassador for Social Support:

Aug 2023 - May 2024

Mentored 12 first-year students in the School of Science and Engineering, providing academic guidance and connecting them to campus resources.